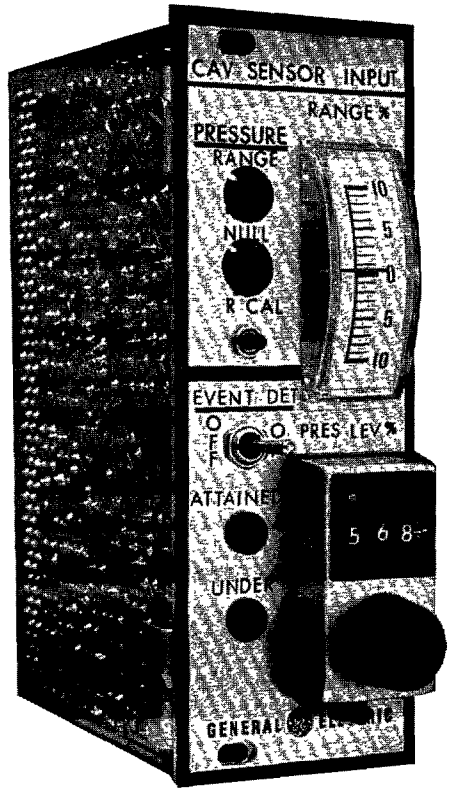
INSTRUCTIONS

PM1000* CONTROL

GEK-40682

CAVITY EVENT DETECTOR

TYPE IC7966



The information contained herein is intended to assist users and dealers in the servicing of control furnished by the General Electric Company. It does not purport to cover all details or variations in equipment or provide for every possible contingency to be met in connection with installation, operation or maintenance.

Should further information be desired or should particular problems arise which are not covered sufficiently for the Purchaser's purpose, the matter should be referred to the machinery manufacturer through his normal service channels.

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GENERAL ELECTRIC

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PM1000* CONTROL

CAVITY EVENT DETECTION

TYPE IC7966

SECTION 1
INTRODUCTION

1.1 PROCESS CONTROL

Process Control is used to refer to a group of functions that is aimed at making the process more consistent and more controllable. Some process control functions are related to increased control of the machine. Examples are Programmed Injection and Hydraulic Pressure or Flow Control. Other process control functions relate to a condition in the process and provide a signal or an adjustment to the machine controls. Examples are Cavity Event Detection, Melt Viscosity Control and Cavity Pressure Regulation.

1.2 SEQUENCE CONTROL

Process Control adds to the basic controls of time, temperature and sequence. All functions in this book are described as being connected to a PM1000 sequence control.

1.3 POWER SUPPLY

The process control functions are designed to connect to the sequence control power supply.

1.4 SEQUENCE CONTROL INTERFACE

The PM1000 process control logic is compatible with the PM1000 Flo-Thru sequence control logic. Hence, wiring is the only required interface between the two controls.

1.5 METERLESS TROUBLESHOOTING

One of the successes of the PM1000 Flo-Thru control is the logic indicating lights that permit troubleshooting without using a meter. This same approach is carried into the PM1000 Process Control with indicating lights and card front meters. The card front meters also permit installation adjustments to be made without an external meter.

1.6 OFF-ON SWITCHES

Each process control function has its individual OFF-ON switch, rather than a master process control OFF-ON switch. This provides complete flexibility over the combinations of functions used. When a function is in the OFF position, the indicating lights are permitted to operate as normal for set-up reasons; however, output correction or alarm signals are disabled.

1.7 LOW-LEVEL WIRING

The electrical connections to the cavity pressure transducer are low-voltage wires. They must be shielded as indicated by the elementary diagram and kept separate from higher voltage wiring.

1.8 CAUTION FOR REMOVING OR INSERTING CARDS

Power must be removed from the panel prior to either removing or inserting printed-circuit cards. Removing or inserting a card without removing power can damage the card due to transient conditions.

1.9 GENERAL MAINTENANCE

These instructions are detailed in the standard sequence manual GEK-40400. They should be reviewed in conjunction with using this process control instruction book.

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SECTION 2
CAVITY EVENT DETECTION

2.1 WHAT THE CAVITY EVENT DETECTOR DOES

The Cavity Event Detector provides cavity pressure repeatability and greatly reduces both flashed (overfilled) and short shots (underfilled) due to uncontrollable pressure variations on the cavity.

2.2 GENERAL OPERATION

The Cavity Event Detector detects a switching pressure point as the mold cavity fills. This previously determined optimum pressure point ("EVENT") is used to signal the basic machine Flo-Thru sequence logic to switch from boost pressure to holding pressure . (See Functional Diagram, Figure 1.)

2.3 MACHINE DEVICE REQUIRED

This function requires a cavity pressure transducer. Recommended models are listed in Chart 1.

2.4 MODIFICATION TO SEQUENCE CONTROL

The sequence control must contain a holding pressure function. The sequence control wiring must be modified slightly to accept the Flo-Thru contact from the Event Detector card. This Event Detector Flo-Thru contact closes to command the switchover from boost pressure to holding pressure and will remain closed for the duration of Die Lock-up.

The preferred operation is for the Boost Timer to operate as normal, except that its time be increased a few seconds. It then serves as a monitor to limit the Boost Time in the event of an injection malfunction such as a clogged nozzle.

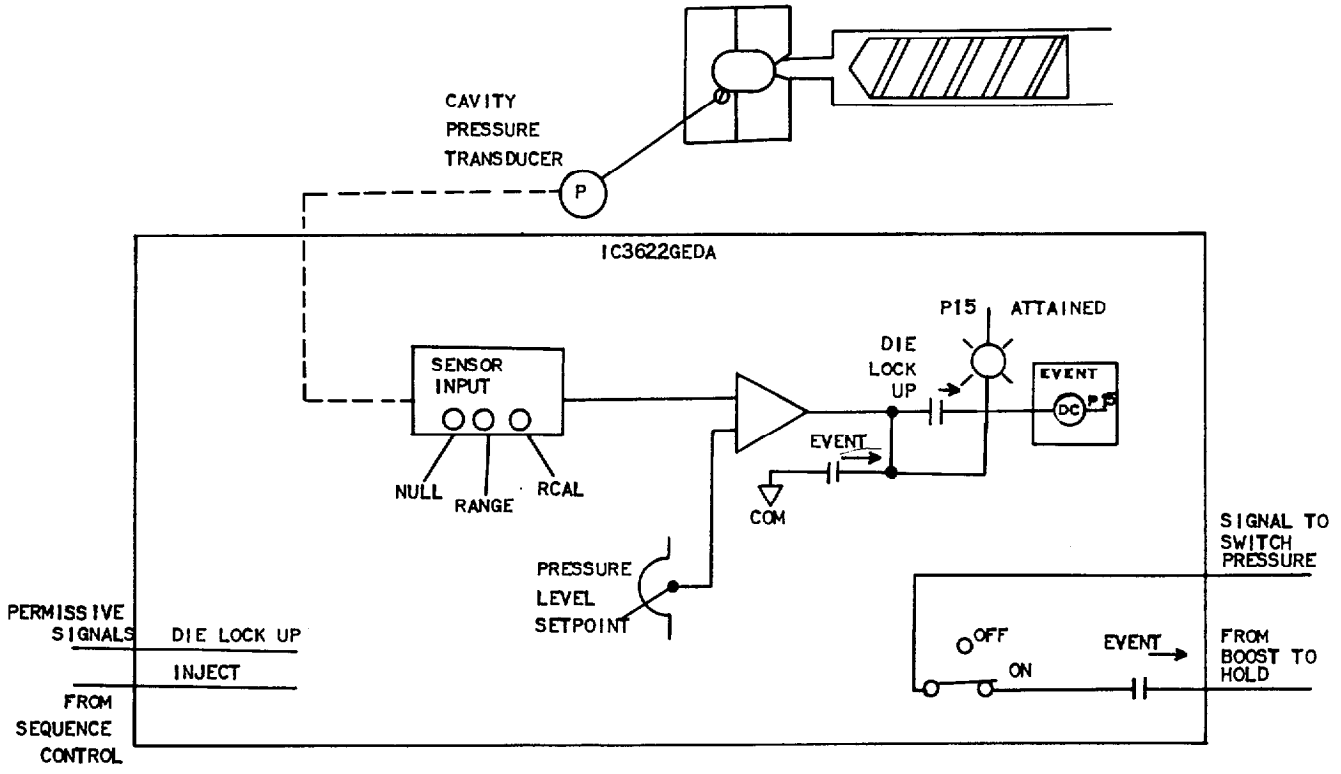


Fig. 1. Functional Diagram for Cavity Event Detection

CAVITY EVENT DETECTION (Continued)

2.6 CONTROL HARDWARE

The circuitry for the Cavity Event Detector is contained on one printed circuit card IC3622GEDA. (Shown by Fig. 2). The GEDA card occupies two PM1000 card slots but only requires a receptacle in the left slot.

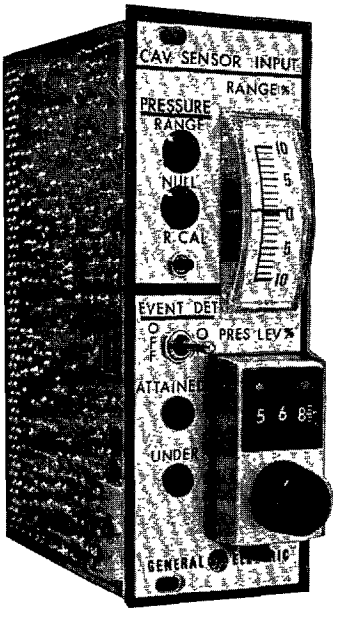


Fig. 2. IC3622GEDA Card

SECTION 3

OPERATOR SET-UP

Listed below are the steps for normal operator set-up. These steps assume the Installation Adjustments have been completed per SECTION 4, and the machine is ready to be put in operation.

Normal Operator Set-Up

1. Set the Event Detector reference at a low number. Turn the Event Detector OFF-ON switch to the ON position.
2. Increase the Boost Time by 3 to 5 seconds greater than the normal time. The Boost Timer then serves as a back-up to the Event Detector.
3. On successive machine cycles, increase the Event Detector Reference until the part is filling properly.

SECTION 4

INSTALLATION ADJUSTMENTS

The instructions below assume the transducer has been connected and the calibration resistor mounted per SECTION 6. These instructions also assume that a Standard Transducer Application per Chart 1 is being used (otherwise, See Note 3 before proceeding).

4.1 NULL (See Figure 3) (See Note 1)

With the machine at rest and the die open, adjust the NULL potentiometer on the GEDA card with a screwdriver until the Sensor Input Meter pointer rests on the zero percent position.

4.2 RANGE (See Figure 3)

Depress the R CAL push button and observe an up-scale pressure indication by the Sensor Input Meter. While holding the push button depressed, adjust the RANGE potentiometer with a screwdriver until the meter pointer rests on the specified calibration percent as specified by the transducer nameplate or data sheet. (See Note 2 if the transducer calibration output is specified in PSI).

Recheck the NULL and then repeat step 4.2. The function is now ready for the Normal Operator Set-Up per SECTION 3.

Note 1. *Mechanical Zero of Meter. The first time a new control or a new card is calibrated, the Sensor Input Meter on the GEDA card should be mechanically zeroed before performing the NULL procedure. To perform this mechanical zero, remove power, remove the card from the panel, and move the tab on the back of the meter until the meter pointer rests at zero. During this procedure the card must be held at about the same relative slope as its position in the panel.*

Note 2: *If the transducer R CAL output is specified in PSI, divide by 200 to obtain percent. For example, suppose the calibration output is 8640 PSI. The desired percent is 8640÷200 = 43.2 percent.*

Note 3. *If the transducer is not being applied per Chart 1, a corrected calibration value must be obtained per the information in SECTION 7.2 before making the RANGE adjustment. If the exception from Chart 1 is because the Pin Size Selector function is being used, review SECTION 8 before proceeding.*

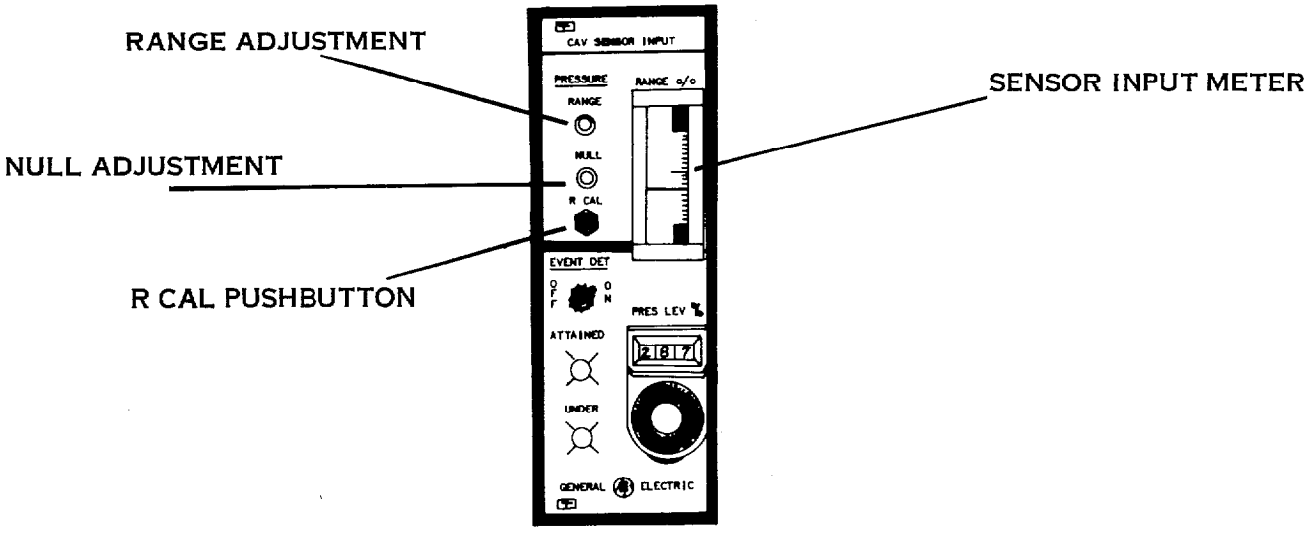


Fig. 3. Installation adjustment devices on GEDA card

SECTION 5

DESCRIPTION OF OPERATION

5.1 TYPICAL ELEMENTARY SHEET

See Fig. 4 (Page 8), which is a reduction of elementary sheet PC-ED-1. It contains several notes referring to installation, adjustments, and operator set-up, which have been expanded in the appropriate sections of this manual.

5.2 INPUT AND OUTPUT SIGNALS

This function requires three inputs, which consist of a cavity pressure transducer and two logic signals. The logic signals are Die Lock-up and Inject.

The major output signal is a Flo-Thru contact that closes when the cavity pressure reaches the Event Reference level. This contact will then remain closed until the end of die lock-up.

Further information on these input and output signals is provided by Fig. 4 (Page 8).

5.3 NORMAL SEQUENCE

A normal sequence is one where the cavity pressure rises properly to trip the Event Detector. The correct input and output signals for such a sequence are described on Fig. 4 (Page 8).

5.4 FAULT SEQUENCE

A fault sequence is one where the cavity pressure does not rise sufficiently as anticipated to trip the Event Detector. The input signals and the resulting output signals for such a sequence are described on Fig. 4 (Page 8).

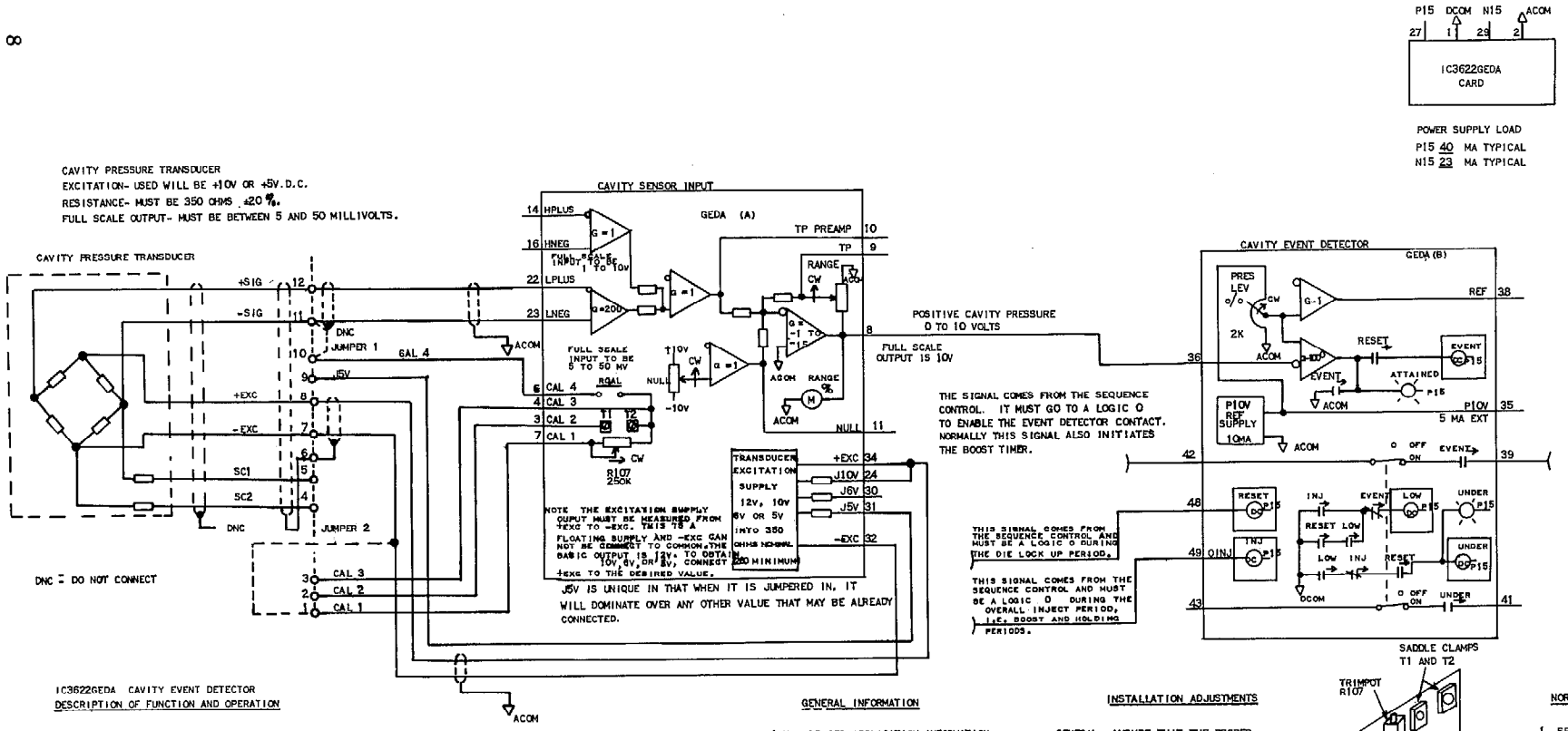


Fig. 4.

IC3622GEDA CAVITY EVENT DETECTOR
DESCRIPTION OF FUNCTION AND OPERATION

THIS CARD CONSISTS OF TWO PORTIONS, A SENSOR INPUT FUNCTION AND AN EVENT DETECTOR FUNCTION. THE SENSOR INPUT PORTION CONDITIONS AND INCREASES A TRANSDUCER OUTPUT TO A WORKING LEVEL VOLTAGE.

THE CAVITY EVENT DETECTOR FUNCTION PERFORMS THE EQUIVALENT OF A PRESSURE SWITCH THAT WILL PROVIDE A SIGNAL WHEN THE MOLD CAVITY PRESSURE REACHES A PRESET LEVEL. THIS SIGNAL IS TYPICALLY USED TO SWITCH THE MOLDING MACHINE FROM THE INJECTION BOOST PRESSURE TO THE INJECTION HOLD PRESSURE.

THE FUNCTION REQUIRES THAT A CAVITY PRESSURE TRANSDUCER BE MOUNTED BEHIND AN UNUSED EJECTOR PIN TO SENSE THE MOLD CAVITY PRESSURE. THIS PRESSURE SIGNAL IS COMPARED BY THE EVENT DETECTOR CARD TO AN ADJUSTABLE REFERENCE ON THE CARD. WHEN THE CAVITY PRESSURE REACHES THIS REFERENCE VALUE, THE CIRCUIT ACTUATES AND AN "ATTAINED" LIGHT ON THE CARD FRONT WILL ILLUMINATE. THIS LIGHT AS WELL AS THE OUTPUT SIGNAL WILL REMAIN UNTIL THE END OF THE MOLD LOCK-UP PERIOD.

THE BOOST TIMER OPERATES AS NORMAL, BUT ITS TIME SETTING MUST BE EXTENDED A FEW SECONDS. IT THEN SERVES AS A BACK UP WHEN THE EVENT DETECTOR LEVEL IS NOT ATTAINED. (SEE FAULT SEQUENCE.)

NORMAL SEQUENCE (EVENT DETECT LEVEL IS ATTAINED)

1. THE MOLD LOCKS UP.
2. THE MACHINE INJECTS IN THE BOOST PRESSURE CONDITION.
3. THE MOLD CAVITY PRESSURE STARTS INCREASING AND REACHES THE EVENT DETECTION REFERENCE VALUE.
4. THE "ATTAINED" LIGHT COMES ON AND THE EVENT DETECTOR OUTPUT CONTACT CLOSES TO SWITCH THE MACHINE TO THE INJECTION HOLDING PRESSURE CONDITION. (THE OFF-ON SWITCH ON THE CARD MUST BE "ON".
5. THE BOOST TIMER TIMES OUT (NO EFFECT).
6. THE INJECT TIMER TIMES OUT TO END THE INJECTION PERIOD.
7. THE MOLD UNLOCKS, THE "ATTAINED" LIGHT GOES OUT AND THE EVENT CONTACT OPENS.

FAULT SEQUENCE (EVENT DETECTION LEVEL NOT ATTAINED)

1. THE MOLD LOCKS UP.
2. THE MACHINE INJECTS IN THE BOOST PRESSURE CONDITION.
3. THE MOLD CAVITY PRESSURE DOES NOT REACH THE EVENT DETECTOR REFERENCE VALVE.
4. THE BOOST TIMER TIMES OUT TO SWITCH THE MACHINE TO THE INJECTION HOLDING PRESSURE.
5. THE INJECT TIMER TIMES OUT TO END THE INJECTION PERIOD.
6. THE "UNDER " LIGHT ON THE CARD FRONT COMES ON TO INDICATE THAT THE EVENT DETECTOR REFERENCE VALUE WAS NOT REACHED AND THE UNDER CONTACT WILL CLOSE.
7. THE MOLD UNLOCKS AND THE "UNDER" LIGHT GOES OUT, AND THE UNDER CONTACT OPENS.

GENERAL INFORMATION

1. TRANSDUCER APPLICATION INFORMATION- SEE SECTION 7
2. TRANSDUCER R-CAL CONNECTION
   A) WHEN TRANSDUCER VENDOR PROVIDES THE RCAL RESISTANCE VALUE JUMPERS 1 AND 2 ARE SHOWN CORRECTLY ABOVE. IN THIS CASE THE CARD IS REMOVED FROM THE PANEL, AN OHMMETER IS CONNECTED TO PINS 7 AND 4 ON THE CARD, AND TRIMPOT R107 IS ADJUSTED FOR THE SPECIFIED RESISTANCE.
   B) WHEN THE TRANSDUCER VENDOR PROVIDES THE R-CAL RESISTOR AS A COMPONENT. JUMPER 1 IS SHOWN CORRECTLY ABOVE. JUMPER 2 MUST BE MOVED FROM CAL 1 TO CAL 2. THE CARD MUST BE REMOVED FROM THE PANEL AND THE RESISTOR MOUNTED BETWEEN SADDLE CLAMPS T1 AND T2 PROVIDED ON THE CARD.
   C) WHEN THE TRANSDUCER VENDOR PROVIDES THE R-CAL RESISTOR INTERNAL TO THE TRANSDUCER. JUMPER 1 MUST BE MOVED TO CONNECT SC1 AND CAL 4. JUMPER 2 MUST BE MOVED TO CONNECT SC2 AND CAL 3.
3. TRANSDUCER EXCITATION VOLTAGE THE EXCITATION VOLTAGE IS SHOWN FOR +10V. TO CHANGE THIS TO +5V, ADD A JUMPER FROM +EXC TO J5V (THE WIRE DOES NOT HAVE TO BE REMOVED FROM PIN 34 TO 24 ON THE CARD RECEPTACLE). USE +10V EXCITATION ON TRANSDUCER THAT LIST +12V MAXIMUM AND USE +5V EXCITATION ON TRANSDUCERS THAT LIST +6V MAXIMUM.

INSTALLATION ADJUSTMENTS

GENERAL- INSURE THAT THE PROPER EXCITATION VOLTAGE IS CONNECTED TO THE CAVITY PRESSURE TRANSDUCER PER THE TRANSDUCER DATA SHEET.

(SEE GENERAL INFORMATION ON THIS SHEET)

1. NULL- WITH THE MACHINE AT REST (E.G. DIE OPEN), THE "NULL" POTENTIOMETER IS ADJUSTED BY A SCREWDRIVER UNTIL THE METER POINTER RESTS ON THE ZERO POSITION.
2. RANGE- R-CAL DEPRESSING THE "R CAL" PUSHBUTTON SUBSTITUTES A RESISTANCE FOR THE CAVITY PRESSURE TRANSDUCER INPUT. THE RESISTANCE IS PRE-SELECTED TO MATCH THE OUTPUT OF THE TRANSDUCER BEING USED. WITH THE "R CAL" PUSHBUTTON DEPRESSED, THE "RANGE" POTENTIOMETER IS ADJUSTED BY SCREWDRIVER UNTIL THE METER POINTER RESTS ON THE DESIRED RCAL OUTPUT AS SPECIFIED BY THE TRANSDUCER DATA SHEET, AND THE INFORMATION BELOW.

IF TRANSDUCER R-CAL OUTPUT IS SPECIFIED IN PSI - DIVIDE THE PSI BY 200 AND THE RESULT IS THE DESIRED PERCENT FOR STEP 2.

IF TRANSDUCER R-CAL OUTPUT IS SPECIFIED IN PERCENT - USE THE PERCENT DIRECTLY FOR STEP 2.

NORMAL OPERATOR SET-UP

1. SET THE EVENT DETECTOR REFERENCE AT A LOW NUMBER, TURN THE EVENT DETECTOR OFF-ON SWITCH TO THE "ON" POSITION.
2. INCREASE THE BOOST TIME BY 3 TO 5 SECONDS GREATER THAN THE NORMAL TIME. THE BOOST TIMER THEN SERVES AS A BACK-UP TO THE EVENT DETECTOR.
3. INCREASE THE EVENT DETECTOR REFERENCE UNTIL THE PART IS FILLING PROPERLY.

PROPRIETORY INFORMATION OF GE CO+

REV. 3	REV. 2	REV. 1 12-10-74	APPROVALS	GENERAL ELECTRIC DRIVE SYSTEMS SALEM, VA. U
REV. 4	ISSUED	PRINT	FIRST MADE FOR REQ. PC STANDARD LINE	ELEMENTARY DIAGRAM EVENT DETECTION
REV. 5	MADE BY DEBBIE LEE		I.C. NO. 7966	CONT. ON SH.

SECTION 6

CONNECTING THE TRANSDUCER AND MOUNTING THE CALIBRATION RESISTOR

6.1 GENERAL

The information below describes how to connect a cavity pressure transducer to the control terminal board for each of the three R-Cal Approaches, and also for an excitation voltage of +10 volts d-c or +5 volts d-c.

An explanation of the R-Cal Approaches is as follows:

It is not practical to apply a known force to the cavity pressure transducer to calibrate it into the circuit. Therefore, a technique called R-Cal calibration must be used. This technique consists of shunting a specified resistance across a portion of the transducer to generate a specified signal for calibration. This resistor is provided by the transducer vendor in one of the following three approaches:

Approach 1 — The R-Cal resistor is built internal to the transducer.

Approach 2 — The transducer vendor supplies the R-Cal resistor as a loose component.

Approach 3 — The transducer vendor provides the R-Cal resistance value only.

6.2 CONNECT FOR PROPER TRANSDUCER EXCITATION VOLTAGE

It is recommended that +10 volts d-c be used for the transducer excitation voltage since it is available at the customer terminal board between points +EXC and -EXC. If +10 volts exceeds the maximum voltage specified on the transducer nameplate or data sheet, +5 volts is available by adding jumper J3 from +EXC to J5V on the customer terminal board. This assumes of course that the GEDA card is wired to the customer terminal board per Fig. 5, Page 10.

6.3 CONNECT FOR PROPER TRANSDUCER CALIBRATION

A. If **Calibration Approach 1** is to be used, connect the transducer to the terminal board per Fig. 6, Page 10. Note J1 and J2 connections.

B. If **Calibration Approach 2** is to be used, connect the transducer to the terminal board per Fig. 7, Page 11. Note J1 and J2 connections.

C. If **Calibration Approach 3** is to be used, connect the transducer to the terminal board per Fig. 8, Page 11.

6.4 MOUNT THE CALIBRATION RESISTOR

A. If **Calibration Approach 1** is to be used, the calibration resistor is internal to the transducer and no mounting is required.

B. If **Calibration Approach 2** is to be used, mount the calibration resistor as follows:
With power off, remove the GEDA card from the panel and mount the R-Cal resistor between saddle clamps T1 and T2 on the GEDA card. (See Fig. 9, Page 12.) Reinsert the GEDA card into the panel.

C. If **Calibration Approach 3** is to be used, mount the calibration resistor as follows:
With power off, remove the GEDA card from the panel. Connect an ohmmeter to pins 7 and 4 on the card and adjust trimpot R107 (see Fig. 9) with a screwdriver for the correct R-Cal resistance as specified by the transducer nameplate or data sheet. (The calibration error will roughly equal the error of the meter being used. An ordinary volt-ohmmeter will provide sufficient repeatability, but if absolute accuracy is desired, a more accurate meter should be used.) Reinsert the GEDA card into the panel.

CONNECTING THE TRANSDUCER AND MOUNTING THE CALIBRATION RESISTOR (Continued)

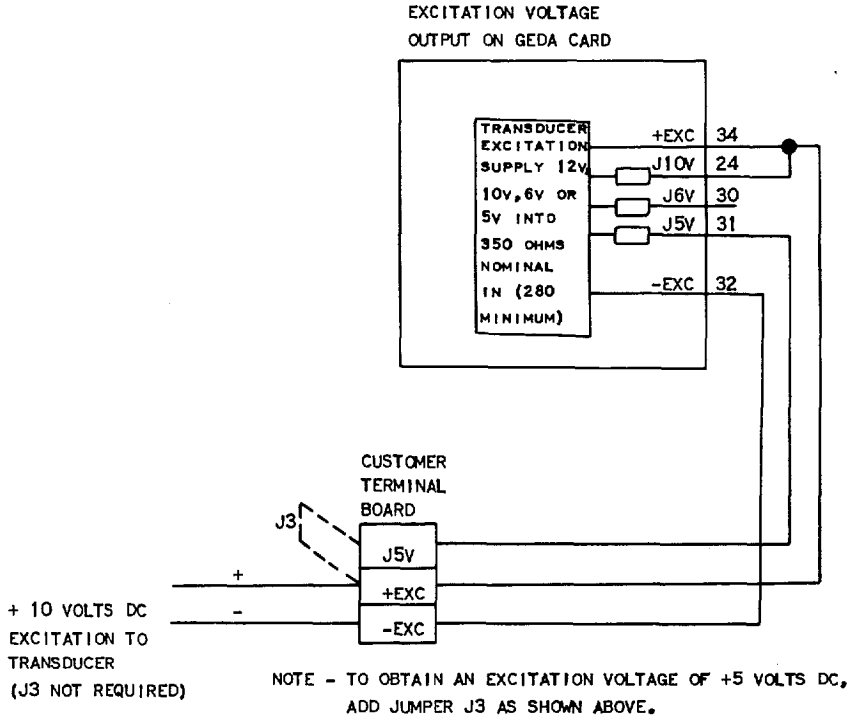


Fig. 5. Transducer excitation voltages of +10 volts d-c or +5 volts d-c

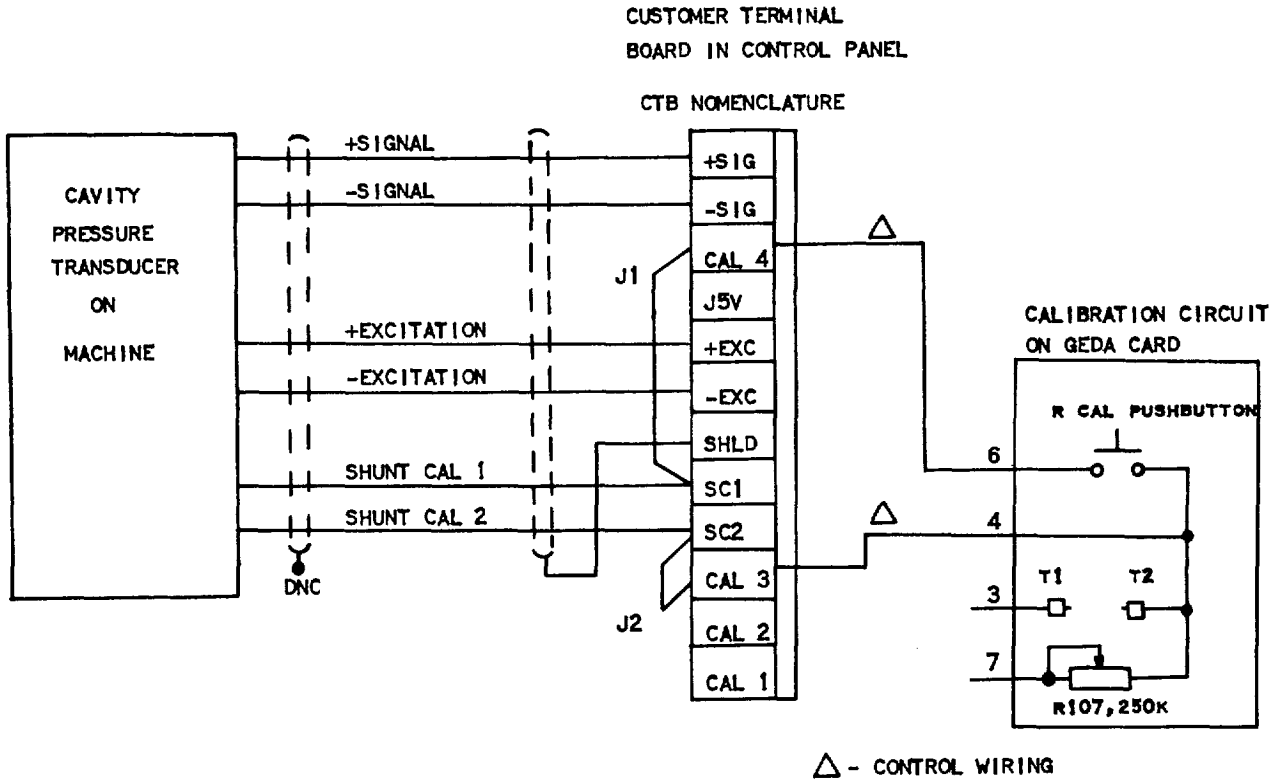


Fig. 6. Terminal board connection when Approach 1 used for transducer R-Cal

CONNECTING THE TRANSDUCER AND MOUNTING THE CALIBRATION RESISTOR (Continued)

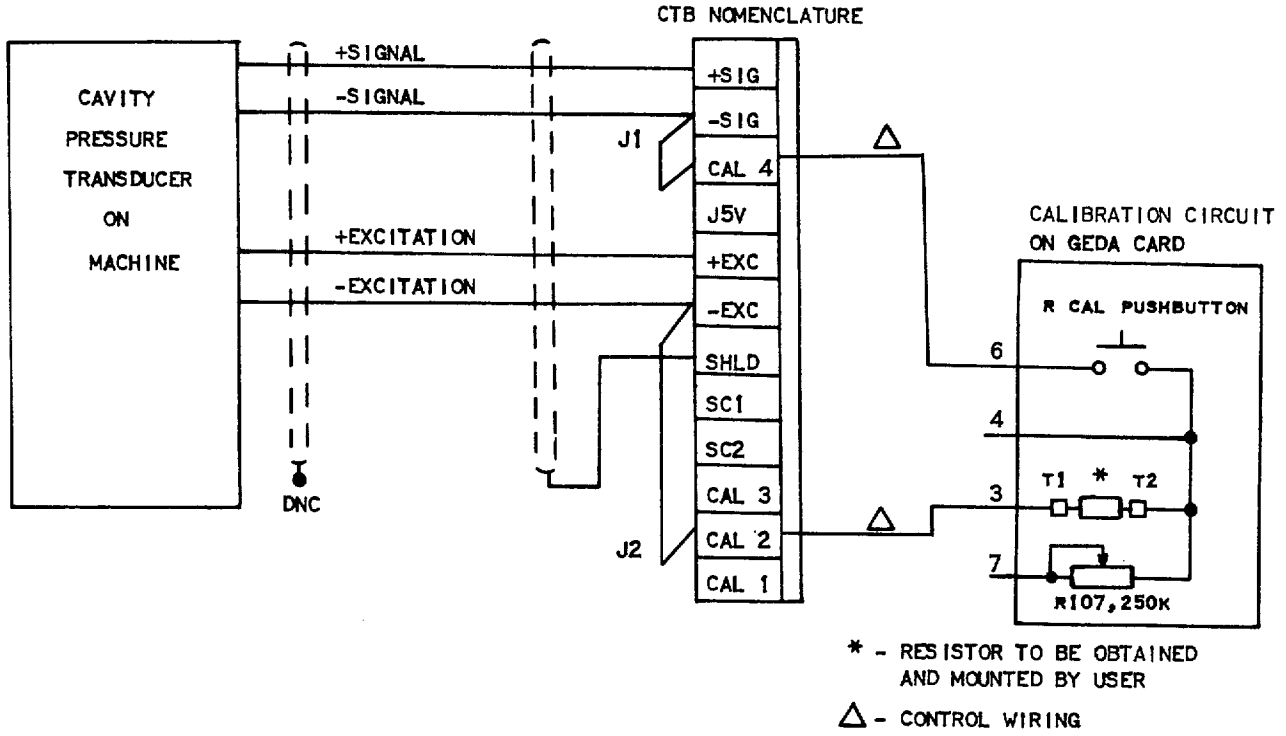


Fig. 7. Terminal board connection when Approach 2 is used for transducer R-Cal

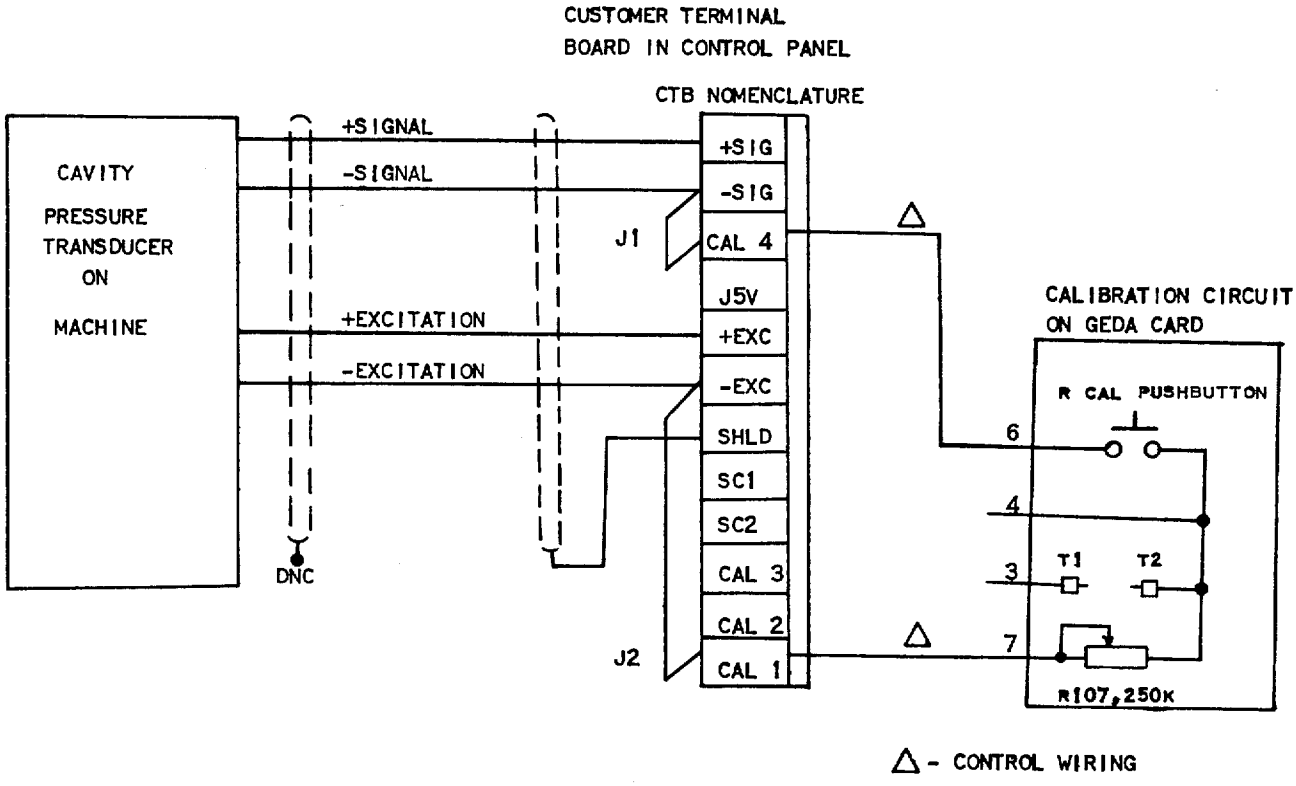


Fig. 8. Terminal board connection when Approach 3 is used for transducer R-Cal

CONNECTING THE TRANSDUCER AND MOUNTING THE CALIBRATION RESISTOR (Continued)

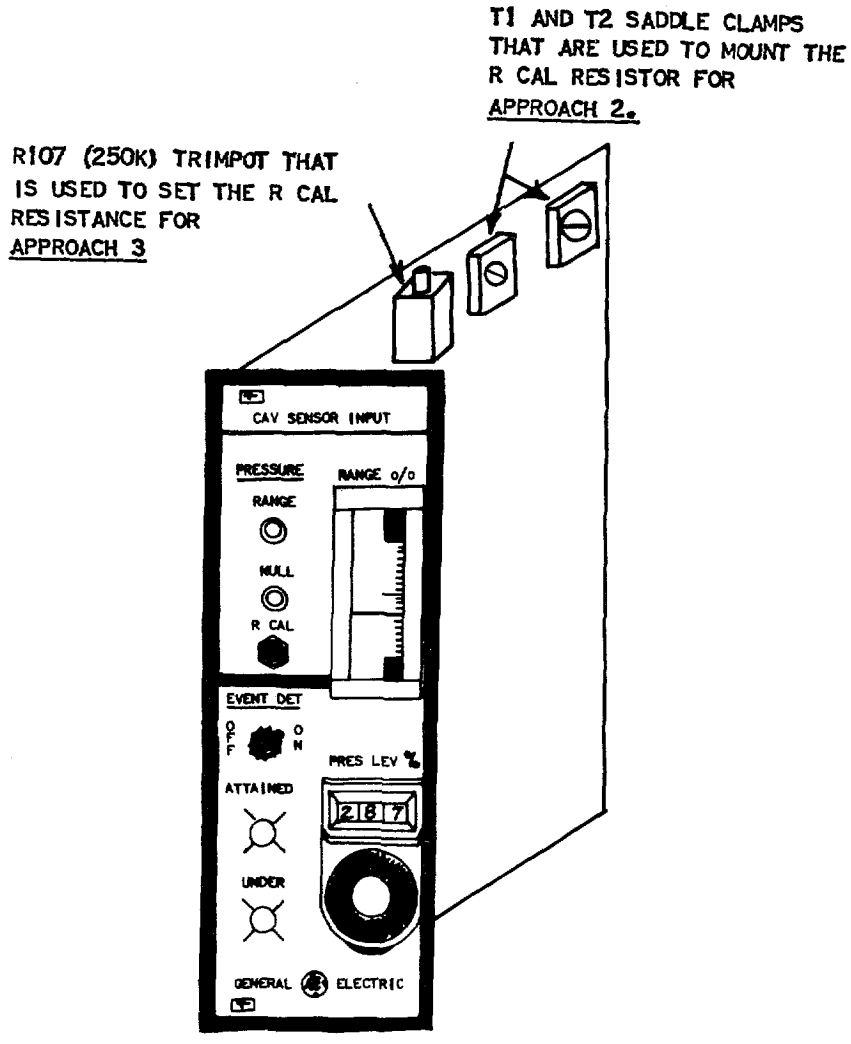


Fig. 9. Location of R-Cal trimpot R107, and R-Cal Saddle Clamps T1 and T2 on GEDA Card

SECTION 7

APPLICATION AND SELECTION OF CAVITY PRESSURE TRANSDUCERS

7.1 RECOMMENDED CAVITY PRESSURE TRANSDUCER AND EJECTOR PIN SIZES

CHART 1 below tabulates recommended pressure transducers and the ejector pin size behind which they are to be used for the PM1000 Event Detector.

CHART 1. Recommended Transducer Applications

Vendor	Transducer Model	Transducer Rating (Pounds of Force)	Use for 1/8" Pin	Use for 3/16" Pin	Use for 1/4" Pin	Use for 3/8" Pin	Use for 1/2" Pin
Dynisco	FT444 - 2.5C	250	X				
Dynisco	FT444 - 5.5C	550		X			
Dynisco	FT444 - 1M	1000			X		
Dynisco	FT444 - 2.2M	2200				X	
Sensotec	MP-300	300	X				
Sensotec	MP-500	500		X			
Sensotec	MP-1000	1000			X		
Sensotec	MP-2000	2000				X	
Sensotec	MP-4000	4000					X
CPI	405	400	X				
CPI	406	2000			X		
CPI	407	8000					X

X = *Indicates recommended applications.*

This application chart matches the transducer to the pin size to allow for a maximum cavity pressure of 20,000 PSI. Although not recommended, information is provided in SECTION 7.2 to apply a transducer in exception to the above chart. If the Ejector Pin Size Selector and Pressure Digital Readout Option is implemented, the transducer application is to be from Chart 2 (page 15) instead of this chart.

APPLICATION AND SELECTION OF CAVITY PRESSURE TRANSDUCERS (Continued)

7.2 NON-STANDARD APPLICATIONS

A transducer can be applied behind an ejector pin size other than the nominal size indicated by CHART 1, if special considerations are given in two areas. (1) not to overload the transducer and (2) correcting the transducer calibration value. If the normal Installation Adjustment procedure is followed and (2) is ignored, the GEDA card will operate satisfactorily, however, the reading on the Sensor Input Meter on the GEDA card cannot be directly interpreted in PSI. (i.e. the 100% = 20,000 PSI relationship is lost). This loss is not so important when the Event Detector is used alone. However, it does become important when a Cavity Regulator function is used along with the Event Detector, and the operator is attempting to set PSI bands about the peak cavity pressure.

The recommended approach is to apply each transducer to only its nominal pin size per CHART 1 unless the Pin Size Selector Option (see SECTION 8) is included in the control package.

If the situation dictates that the transducer must be used behind pin sizes other than the nominal pin sizes, the following statements will be helpful.

A. If the pin size is larger than the nominal pin size, care must be taken not to overload and damage the cell. The maximum PSI rating can be calculated by dividing the cell rating (pounds) by the area of the ejector pin end (inches ^{2}) exposed to the plastic.

B. If the pin size used is smaller than the nominal pin size, control sensitivity diminishes, and the temperature drift of the transducer signal increases.

C. Correct the transducer calibration value (supplied on the transducer nameplate or data sheet) by multiplying it by the

$$\left(\frac{\text{Diameter of Nominal Pin Size in Inches}}{\text{Diameter of Actual Pin Size in Inches}}\right)^2$$

Record the corrected value. Certain transducers and their nominal pin sizes are indicated in CHART 1.

D. Supply the corrected calibration value of C to the set-up man for his use in making Installation Adjustments.

7.3 TRANSDUCER ELECTRICAL SPECIFICATIONS

If a transducer is to be used that does not appear in CHART 1, the following electrical criteria can be used to determine if the transducer is electrically compatible with the GEDA card.

Excitation:	+10 volts d-c or +5 volts d-c
Output:	The full scale output must be between 5 and 50 millivolts d-c
Resistance:	Must be 350 ohms ±20%
Temperature Drift:	Typical transducer specifications are 0.2% full scale/°F on NULL and 0.02% reading/°F on RANGE.
Temperature Range:	User must determine this based on his Mold Operating Temperature.
Correct Pin Size:	Apply the transducer to the correct pin size per CHART 1.

7.4 BENEFITS OF STANDARDIZING ON ONE MAKE OF TRANSDUCER

The GEDA card is equipped to be compatible with several transducer makes and three calibration approaches. The necessary signals to provide this flexibility are wired to the Customer terminal board. Hence, a user can convert from one transducer make to another by making the proper connections at the Customer terminal board.

However, it is recommended that each user standardize on one make of transducer and one calibration approach. The benefits are as follows:

A. The connections of the transducer leads to the terminal board become standard, thus greatly simplifying the connection of a new transducer.

B. The calibration resistor may not have to be replaced on the GEDA card when a new transducer is installed, because most transducers of the same make use the same R-Cal resistor.

C. Due to "A", it can be recommended that a quick-disconnect plug for the transducer be mounted on the machine. It should be noted that most transducer vendors will provide transducers with a plug attached.

D. If suggestions A, B, and C are implemented, the procedure for changing a transducer can reduce to unplugging the old, plugging in the new and performing the two adjustments (NULL and RANGE) on the Event Detector Card.

SECTION 8
OPTIONAL FUNCTION
EJECTOR PIN SIZE SELECTOR AND PRESSURE DIGITAL READOUT

The basic Cavity Event Detector essentially requires that each cavity pressure transducer be matched to one size of ejector pin.

The Pin Size Selector and Digital Readout Option provides a means of using the same transducer behind different size ejector pins. This option provides for calibration of the transducer directly in PSI using the digital readout. This calibration is maintained for different size ejector pins by selecting the pin size being used with the selector switch.

When this option is implemented, the following steps must be added to the Installation Adjustment procedure listed in SECTION 4.

Add Step Preliminary: Rotate the Pin Size Selector Switch to the NOMINAL pin size per Chart 3 on Fig. 10, Page 16.

Change Step A: Adjust the NULL for 0 on the Digital Readout.

Change Step B: Adjust the RANGE for the calibration PSI on the Digital Readout.

Add Step C: Rotate the Pin Size Selector Switch to the ACTUAL pin size in the mold being used.

If the transducer calibration output is specified in percent, multiply by 20,000 to get PSI.

The proper transducer applications and additional information for this option are provided on Chart 2 below.

8.1 CAVITY PRESSURE TRANSDUCER APPLICATIONS FOR EJECTOR PIN SIZE SELECTOR AND PRESSURE DIGITAL READOUT OPTION.

CHART 2 below tabulates recommended pressure transducers and the ejector pin size behind which they are to be used for the PM1000 Event Detector when the pin size selection and pressure digital readout option is implemented.

CHART 2. Recommended Transducer Applications for Pin Size Selection and Pressure Digital Readout Option.

Vendor	Transducer Model	Transducer Rating (Pounds of Force)	Use for 1/16" Pin	Use for 3/32" Pin	Use for 1/8" Pin	Use for 3/16" Pin	Use for 1/4" Pin	Use for 5/16" Pin	Use for 3/8" Pin	Use for 1/2" Pin
Dynisco	FT444-2.5C	250	x	x	⊗	Max 8,900 PSI				
Dynisco	FT444-IM	1000				x	⊗	Max 12,800PSI	Max 8,900PSI	Max 5,000 PSI
Sensotec	MP-300	300	x	x	⊗	Max 8,900 PSI				
Sensotec	MP-1000	1000				x	⊗	Max 12,800PSI	Max 8,900PSI	Max 5,000 PSI
CPI	405	400	x	x	⊗	Max 8,900 PSI				
CPI	406	2000				x	⊗	Max 12,800PSI	Max 8,900PSI	Max 5,000 PSI

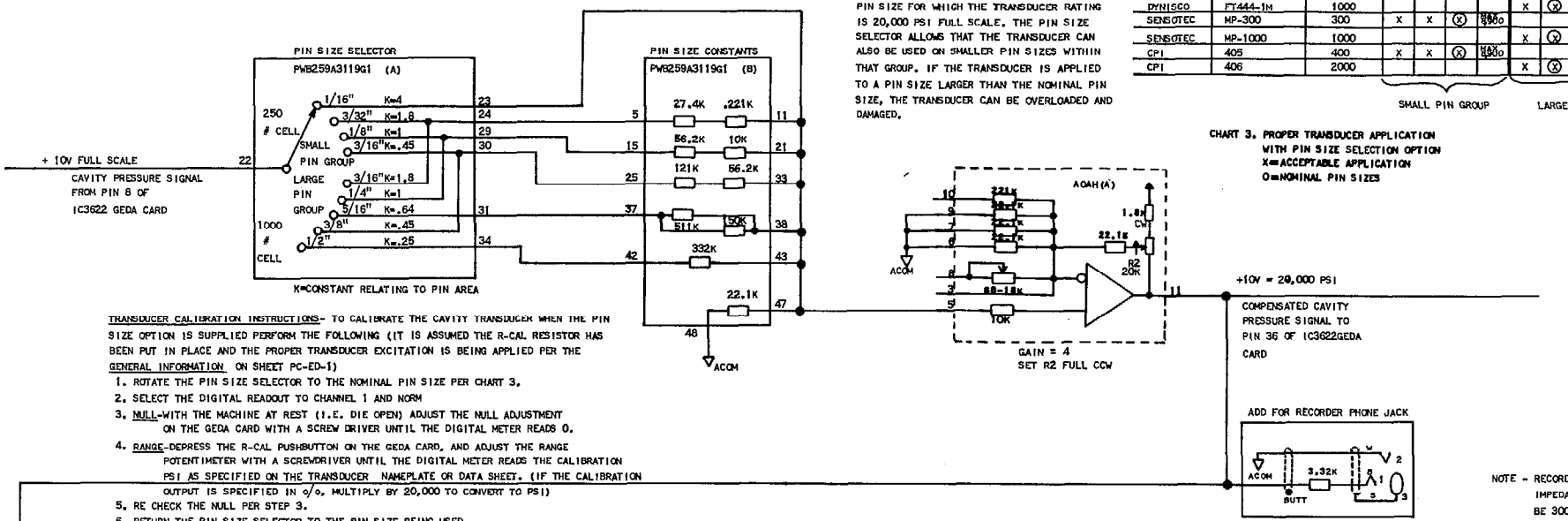
x = *Indicates recommended applications.* ⊗ = *Nominal pin size i.e. pin size which matches transducer 20,000 PSI rating.*
NOTE: *The maximum (Max) PSI values are a combination of the transducer rating and a control saturation voltage.*

TRANSDUCER APPLICATION- THE CORRECT TRANSDUCER APPLICATIONS FOR VARIOUS PIN SIZES ARE TABULATED IN THE ADJACENT CHART. THIS CHART IDENTIFIES THE TRANSDUCER NOMINAL PIN SIZE I.E. THE PIN SIZE FOR WHICH THE TRANSDUCER RATING IS 20,000 PSI FULL SCALE. THE PIN SIZE SELECTOR ALLOWS THAT THE TRANSDUCER CAN ALSO BE USED ON SMALLER PIN SIZES WITHIN THAT GROUP. IF THE TRANSDUCER IS APPLIED TO A PIN SIZE LARGER THAN THE NOMINAL PIN SIZE, THE TRANSDUCER CAN BE OVERLOADED AND DAMAGED.

VENDOR	TRANSDUCER MODEL	TRANSDUCER RATING (POUNDS OF FORCE)	SMALL PIN GROUP: USE FOR 1/16" PIN	USE FOR 3/32" PIN	USE FOR 1/8" PIN	USE FOR 3/16" PIN	LARGE PIN GROUP: USE FOR 3/16" PIN	USE FOR 1/4" PIN	USE FOR 5/16" PIN	USE FOR 3/8" PIN	USE FOR 1/2" PIN
DYNISCO	FT444 -2,5C	250	X	X	⊗	MAX 6,500					
DYNISCO	FT444-1M	1000					X	⊗	MAX 12,800	MAX 8,900	MAX 5000
SENSOTEC	MP-300	300	X	X	⊗	MAX 6,500					
SENSOTEC	MP-1000	1000					X	⊗	MAX 12,800	MAX 8,900	MAX 5000
CPI	405	400	X	X	⊗	MAX 6,500					
CPI	406	2000					X	⊗	MAX 12,800	MAX 8,900	MAX 5000

SMALL PIN GROUP — LARGE PIN GROUP

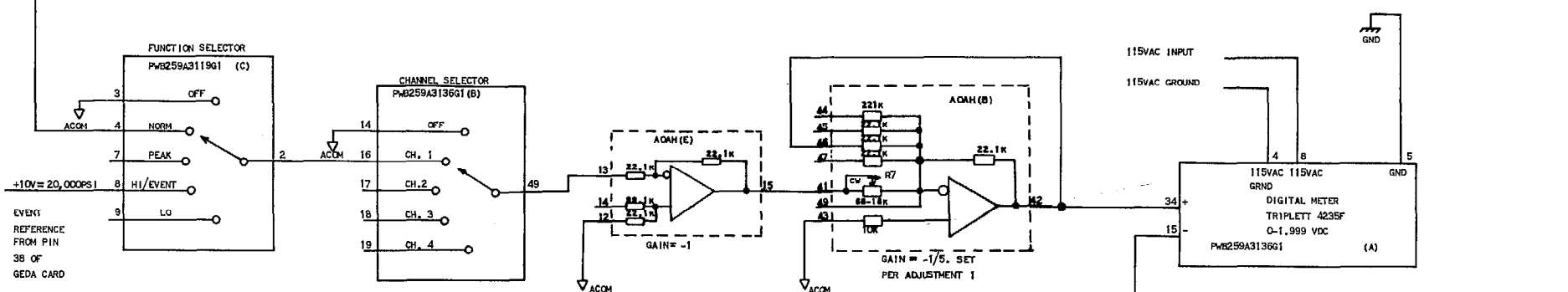
CHART 3. PROPER TRANSDUCER APPLICATION WITH PIN SIZE SELECTION OPTION
X=ACCEPTABLE APPLICATION
O=NOMINAL PIN SIZES



TRANSDUCER CALIBRATION INSTRUCTIONS- TO CALIBRATE THE CAVITY TRANSDUCER WHEN THE PIN SIZE OPTION IS SUPPLIED PERFORM THE FOLLOWING (IT IS ASSUMED THE R-CAL RESISTOR HAS BEEN PUT IN PLACE AND THE PROPER TRANSDUCER EXCITATION IS BEING APPLIED PER THE GENERAL INFORMATION ON SHEET PC-ED-1)

1. ROTATE THE PIN SIZE SELECTOR TO THE NOMINAL PIN SIZE PER CHART 3.
2. SELECT THE DIGITAL READOUT TO CHANNEL 1 AND NORM
3. NULL-WITH THE MACHINE AT REST (I.E. DIE OPEN) ADJUST THE NULL ADJUSTMENT ON THE GEDA CARD WITH A SCREW DRIVER UNTIL THE DIGITAL METER READS 0.
4. RANGE-DEPRESS THE R-CAL PUSHBUTTON ON THE GEDA CARD, AND ADJUST THE RANGE POTENTIMETER WITH A SCREWDRIVER UNTIL THE DIGITAL METER READS THE CALIBRATION PSI AS SPECIFIED ON THE TRANSDUCER NAMEPLATE OR DATA SHEET. (IF THE CALIBRATION OUTPUT IS SPECIFIED IN o/o, MULTIPLY BY 20,000 TO CONVERT TO PSI)
5. RE CHECK THE NULL PER STEP 3.
6. RETURN THE PIN SIZE SELECTOR TO THE PIN SIZE BEING USED.
7. THE CALIBRATION IS NOW COMPLETE.

NOTE - RECORDER INPUT IMPEDANCE SHOULD BE 300K OR GREATER TO PREVENT LOADING ERROR DUE TO 3.32K ISOLATION RESISTOR.

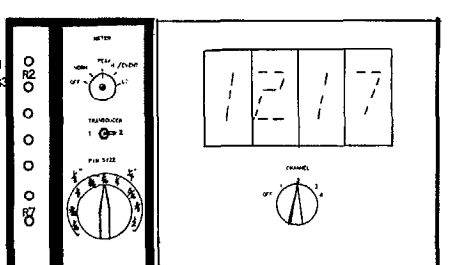


ADJUSTMENT 1 - PROCEDURE FOR CALIBRATION OF DIGITAL METER READOUT

1. SELECT CHANNEL 1 ON METER CHANNEL SELECTOR
2. SELECT HI/EVENT ON METER FUNCTION SELECTOR
3. SET EVENT REFERENCE AT 999
4. ADJUST R7 ON AOAH CARD FOR 1998 ON DIGITAL METER

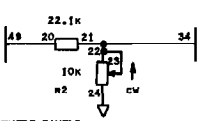
HARDWARE REQUIRED

METER-PWB259A3136G1
PIN SELECTOR-PWB259A3119G1
OPERATIONAL AMPLIFIER CARD-IC3600AOAH
SPECIAL METER RECEPTACLE-PWB259A3136G3



IC3600AOAH PWB259A3119G1 — PWB259A3136G1
1 SLOT 2 SLOTS — 6 SLOTS

ON SOME PANELS, A RESISTANCE DIVIDER WAS USED INSTEAD OF THE AOAH(E) AND AOAH(B) CIRCUITS. THIS DIVIDER WAS AS SHOWN BELOW



ON THESE PANELS SUBSTITUTE "R2 ON SCBN CARD" INTO STEP 4 OF ADJUSTMENT 1 LISTED AT RIGHT.

REV. 3	REV. 2	REV. 1 12-10-74	APPROVALS	GENERAL ELECTRIC DRIVE SYSTEMS DEPT. SALEM, VA. U.S.A.
REV. 4	ISSUED AUGUST 19, 1974	PRINTS TO	FIRST MADE FOR REQ. STD LINE	ELEMENTARY DIAGRAM EVENT DETECTOR DIGIT. READOUT OPTION (TRIPL METER)
REV. 5	MADE BY PAULA L. POFF		I.C. NO. 7966	CONT. ON SH.

Fig. 10.

SECTION 9

FIELD INSTALLATION/CHECKOUT PROCEDURE

A. GENERAL

This information is provided for the situation where the Event Detector is field installed. This procedure should not be necessary if the function was included in the original equipment.

Procedure	Check When Completed

B. BEFORE APPLYING POWER, GEDA CARD OUT

1. Complete the wiring

a. Complete the wiring for the function per the instructions provided, or per Fig. 4, Page 8, of this manual, or per the equivalent elementary sheet. ☐

2. Obtain the Cavity Transducer

a. Obtain the correct Cavity Pressure Transducer by determining the diameter of the ejector pin to be used and then choosing the transducer per CHART 1, Page 13. ☐

3. Connect the Transducer to the Control Terminal Board and mount the calibration resistor per SECTION 6. ☐

4. Continuity Checks to verify the wiring and transducer connections. (Use an ohmmeter for all steps).

a. Measure 350 ohms ±20% between pins 22 and 23 on the GEDA receptacle. ☐

Procedure	Check When Completed

b. Measure 0 ohms or direct connection from pin 1 on the GEDA receptacle to DCOM. ☐

c. Measure 0 ohms or direct connection from pin 2 on the GEDA receptacle to ACOM. (Measure to DCOM if ACOM is not included in panel.) ☐

d. Measure an open circuit from pin 23 on the GEDA receptacle to DCOM. ☐

e. Measure between 280 and 600 ohms from pin 34 to 32 on the GEDA receptacle. ☐

f. Measure an open circuit when pin 34 on the GEDA receptacle to DCOM. ☐

g. If the transducer is connected per Fig. 6, Page 10, measure a resistance of several thousand ohms but less than 250K between pins 4 and 6 on the GEDA receptacle. (If this reading is an open circuit, the R-Cal resistor may not be included inside the transducer.) ☐

h. If the transducer is connected per Fig. 7, Page 11, measure 262 ohms ±20% between pins 3 and 6 on the GEDA receptacle. ☐

i. If the transducer is connected per Fig. 8, Page 11, measure 262 ohms ±20% between pins 7 and 6 on the GEDA receptacle. ☐

FIELD INSTALLATION/CHECKOUT PROCEDURE (Continued)

Procedure	Check When Completed

C. APPLY POWER, GEDA CARD STILL OUT

1. Check power-supply voltages
   a. Use a voltmeter to verify +15 volts d-c from pin 27 to common on the GEDA receptacle. ☐
   b. Use a voltmeter to verify -15 volts d-c from pin 29 to common on the GEDA receptacle. ☐

D. INSTALLATION ADJUSTMENTS

1. Remove power, insert the GEDA card into the panel, apply power and perform the Installation Adjustments per SECTION 4. ☐

E. OPERATE THE MACHINE, TRANSDUCER OUT

1. Electrical Noise Test

   With the cavity pressure transducer not inserted into the mold, operate the machine through a cycle and verify that the pointer on the Sensor Input Meter on the GEDA card does not move away from 0. (This confirms there is no noise pickup on the transducer wiring.) ☐

F. OPERATE THE MACHINE, TRANSDUCER IN

1. Logic Signal Verification
   a. Preliminary
      1. Turn the EVENT DET switch to OFF. ☐
      2. Set the PRESS LEV setpoint on the GEDA card to 999. ☐
   b. Cycle the molding machine with the mold being filled sufficiently to reach the transducer
      1. Verify that the Sensor Input Meter on the GEDA card indicates that a pressure is being measured. ☐
      2. Verify that the UNDER light on the GEDA card comes on at the end of the overall injection period and goes off at the end of die lock-up. (This confirms the correct logic inputs to pins 48 and 49 of the GEDA card.) ☐

2. Functional Checkout
   a. Preliminary
      1. Turn the EVENT DET switch to ON. ☐
   b. Cycle the molding machine with the mold filling sufficiently to reach the transducer. Probably three machine cycles will be required to verify the following three items:
      1. Lower the PRES LEV setpoint such that the ATTAINED light comes on during cavity fill. ☐
      2. Verify that the ATTAINED light resets at the end of die lock-up. ☐
      3. Verify that when the ATTAINED light comes on, the machine is switched into Holding Pressure. (It may be necessary to extend the Boost Timer to confirm this step.) ☐

The Event Detector checkout is now complete. Refer to SECTION 3 for the normal Operator Set-Up Procedure.

SECTION 10

TROUBLESHOOTING

10.1 POWER SUPPLY CONNECTIONS

A. Event Detector Card IC 3622GEDA

1. P15V on pin 27
2. N15V on pin 29
3. DCOM on pin 1
*4. ACOM on pin 2

***Note:** *For PM1000 panels that do not have an ACOM bus, pin 2 must be connected to DCOM.*

10.2 TROUBLESHOOTING CHART

SYMPTOM	PROBABLE CAUSE	REMEDY
Symptom 1 The GEDA output pressure signal will not null to zero.	Excitation voltage is missing or incorrect.	Missing Power Supply connection to GEDA card. Confirm per Item A under this section.
		Open or grounded excitation voltage wires to the transducer.
	Open or grounded signal wires to the transducer.	Identify and correct fault
	Defective GEDA Card.	Replace card.
	Defective Cavity Pressure Transducer.	Replace Transducer.
	Above instructions have not solved problem.	Perform Steps 4 of the Checkout Procedure under Section 9. Note that power must be off and the GEDA card removed from the panel.
Symptom 2 The GEDA output pressure signal does not change when the R-Cal pushbutton is pushed.	Verify that Symptom 1 does not exist.	If Symptom 1 exists, follow the causes and remedies listed for it.
	The R-Cal circuit is open.	Review Section 6 Transducer Connection and verify completeness of R-Cal wiring.
Symptom 3 The GEDA output pressure signal goes negative when the R-Cal pushbutton is pushed.	Either the signal leads or the excitation leads from the transducer are connected improperly.	Review the transducer data sheet and correct the connections.
Symptom 4 The GEDA output pressure signal does not indicate any pressure in the mold during operation.	Verify the transducer is inserted properly behind the ejector pin.	Insert the transducer fully and secure it.
	Verify the ejector pin is not mechanically sticking or binding.	Free it up.
	Verify the mold is filling completely, and the plastic is reaching the transducer.	Correct machine set up
	Verify the proper transducer is being used for the size ejector pin being used.	See Chart 1 in Section 7 of this manual.
	The transducer is defective.	Replace the transducer.

10.2 TROUBLESHOOTING CHART (Continued)

SYMPTOM	PROBABLE CAUSE	REMEDY
Symptom 5 The GEDA output pressure signal fluctuates wildly before the mold closes up.	Remove the transducer from the mold (do not disturb the electrical connections). If the fluctuations persist, there is electrical noise being picked up by the transducer leads.	Verify that the transducer wiring to the control panel is not in the same conduit or close to motor or other power wiring. Verify that the transducer wiring to the control panel is shielded. The shield is to connect to common in the control panel and must be open at the transducer end.
	The ejector pin is being vibrated or strained during the machine cycle.	Investigate and mechanically resolve.
	Defective GEDA card.	Replace card.
	Intermittent Transducer.	Replace transducer.
Symptom 6 The ATTAINED light comes on properly as the mold is being filled, but the machine does not switch to hold pressure.	The input signal into pin 42 of the GEDA card does not go to 0 volts to permit a 0 volt signal on the Event contact output (pin 39).	Investigate and resolve.
	The Event contact output (pin 39) on the GEDA card does not switch to 0 volts when the ATTAINED light comes on.	Incorrect input on pin 42 (see above). The 0 volt permissive signal to pin 48 of the GEDA card is incorrect. Defective GEDA card. Replace.
	The Event contact output on the GEDA card is not wired correctly to enable the Hold Pressure function in the sequence control.	Verify correct wiring from the GEDA output (pin 39) into the Sequence Control Logic.
	If all the above fails, perform checkout per Section 9-F.	
Symptom 7 When the R-Cal pushbutton is pushed, the GAIN adjust will not increase the GEDA output to the specified per cent.	The R-Cal Resistor or resistance being used is the wrong value.	Measure the value of the resistance being used with an ohmmeter and verify it per the transducer data sheet.
	The transducer full scale output is less than 5 millivolts.	Increase the transducer excitation to +10 volts. See Section 6-B.

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